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MAT251_TW | Chapter 8: Sequences and Series (Continued)

Started on	Sunday, August 8, 2021, 4:19 PM
State	Finished
Completed on	Sunday, August 8, 2021, 4:31 PM
Time taken	12 mins 30 secs

Question **1**

Complete

Not graded

If you estimate

$$S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2^n}$$

with

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \frac{1}{32} - \frac{1}{64} + \frac{1}{128} = .3359375\ldots,$$

what is an estimate of the error involved?

Select one:

- a. 1/2
- b. 1/8
- c. 1/128
- d. 1/256

If your approximation of an alternating series uses *m* terms, then the error is less than a_m+1. You've used 7 terms; therefore, the error is less than a_8, which is $\frac{1}{2^8} = \frac{1}{256}$.

The correct answer is: 1/256

Question **2**

Complete

Not graded

How many terms of $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$ do you need to sum in order to be sure your approximation of *S* is accurate to 2 decimal places?

Select one:

- a. 1000
- b. 3
- c. 10
- d. 9

The error $|S - S_m|$ is always less than a_{m+1}.

$$a_{10} = \frac{1}{1000} = 0.001,$$

Because for this problem the error is less than 0.001 when $m + 1 = 10$ or when $m = 9$. But if you have an error of less than 0.001, you are guaranteed 2 decimal places of accuracy. Thus if you sum 9 terms of this alternating series, you are guaranteed 2 decimal places of accuracy.

The correct answer is: 9

Question **3**

Complete

Not graded

What does the alternating series test say about $\sum_{n=1}^{\infty} \frac{(-n)^{n+1}}{n^2+1}$?

Select one:

- a. Nothing, this is not an alternating series and therefore the alternating series test does not apply.
- b. The series diverges because the a_n are not all positive.
- c. The series diverges because $\lim_{n\to\infty} a_n$ is not 0.
- d. The series converges.

This is an alternating series with $a_n = \frac{n^{n+1}}{n^2+1}$. Check the alternating series test:

I. $\frac{n^{n+1}}{n^2+1}$ is positive.

II. $\frac{n^{n+1}}{n^2+1}$ is not decreasing.

III. $\lim_{n\to\infty} \frac{n^{n+1}}{n^2+1} = \infty$.

The second and third conditions are not met; therefore, this series diverges by the alternating series test.

The correct answer is: The series diverges because $\lim_{n\to\infty} a_n$ is not 0.

Question **4**

Complete

Not graded

Does the series

$$\frac{1}{2}-\frac{1}{4}+\frac{1}{6}-\frac{1}{8}+\frac{1}{10}-\frac{1}{12}+\ldots$$

converge or diverge?

Select one:

- a. The series diverges because the a_n are not all positive.
- b. The series diverges because the a_n are not decreasing.
- c. The series diverges because the $\lim_{n\to\infty} a_n$ is not 0.
- d. The series converges.

This is an alternating series with $a_n = \frac{1}{2n}$. Check the alternating series test:

I. $\frac{1}{2n}$ is positive.

II. $\frac{1}{2n}$ is decreasing.

III. $\lim_{n\to\infty} \frac{1}{2n} = 0$.

All three conditions are satisfied; therefore, this series converges by the alternating series test.

The correct answer is: The series converges.

Question **5**

Complete

Not graded

Does the series

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \frac{1}{32} - \frac{1}{64} + \ldots$$

converge or diverge?

Select one:

- a. The series converges.
- b. The series diverges because the a_n are not decreasing.
- c. The series diverges because $\lim_{n\to\infty} a_n$ is not 0.
- d. Nothing, this is not an alternating series and therefore the alternating series test does not apply.

This is an alternating series with $a_n = \frac{1}{2^n}$. Check the alternating series test:

I. $\frac{1}{2^n}$ is positive.

II. $\frac{1}{2^n}$ is decreasing.

III. $\lim_{n\to\infty} \frac{1}{2^n} = 0$.

All the conditions are met; therefore, this series converges by the alternating series test.

The correct answer is: The series converges.

Question **6**

Complete

Not graded

What does the alternating series test say about \(\displaystyle\sum\limits_{n=1}^ \infty \frac{(-1)^{n+1}n}{10n+1} \)?

Select one:

- a. The alternating series test does not give any information about the convergence or divergence of the series.
- b. The series diverges because the \(a_n \) are not all positive.
- c. The series diverges because the \(a_n \) are not decreasing and \ (\displaystyle\lim_{n \to \infty} a_n \) is not 0.
- d. The series converges.

This is an alternating series with $a_n = \frac{n}{10n+1}$. Check the alternating series test:

I. $\frac{n}{10n+1}$ is positive.

II. $\frac{n}{10n+1}$ is not decreasing.

III. $\lim_{n\to\infty} \frac{n}{10n+1} = \frac{1}{10}$.

The second and third conditions are not met; therefore, this series diverges by the alternating series test.

The correct answer is: The series diverges because the \(a_n \) are not decreasing and \(\displaystyle\lim_{n \to \infty} a_n \) is not 0.

Question **7**

Complete

Not graded

Does \(\displaystyle\sum\limits_{n=1}^ \infty \frac{(-1)^{n+1}(n^2+2n+1)}{2n^2+3n+1} \) converge or diverge?

Select one:

- a. The series diverges because the \(a_n \) are not decreasing.
- b. The series diverges because the \(a_n \) are not all positive.
- c. The series diverges because \(\displaystyle\lim_{n \to \infty} a_n\) is not 0.
- d. The series converges.

This is an alternating series with

$$a_n = \frac{n^2+2n+1}{2n^2+3n+1} = \frac{(n+1)^2}{(2n+1)(n+1)} = \frac{n+1}{2n+1}$$

Check the alternating series test:

I. $\frac{n+1}{2n+1}$ is positive.

II. $\frac{n+1}{2n+1}$ is decreasing. $\left(\frac{2}{3} > \frac{3}{5} > \frac{4}{7} > \frac{5}{9} > \cdots\right)$

III. $\lim_{n\to\infty} \frac{n+1}{2n+1} = \frac{1}{2}$.

Because

$$\lim_{n\to\infty} \frac{n^2+2n+1}{2n^2+3n+1}$$

does not equal 0, the alternating series test tells us this series diverges.

The correct answer is: The series diverges because \(\displaystyle\lim_{n \to \infty} a_n\) is not 0.

Question **8**

Complete

Not graded

What does the alternating series test say about \(\displaystyle\sum\limits_{n=1}^ \infty (-1)^{n+1} \)?

Select one:

- a. Nothing, this is not an alternating series and therefore the alternating series test does not apply.
- b. The series diverges because the \(a_n \) are not all positive.
- c. The series diverges because the \(\displaystyle\lim_{n \to \infty} a_n \) is not 0.
- d. The series converges.

This is an alternating series
with $a_n = 1$. Check the alternating
series test:
I. 1 is positive.
II. 1 is not decreasing.
III. $\lim_{n\to\infty} 1 = 1$.
This series does not satisfy the
second and third conditions
and therefore, diverges by the
alternating series test.

The correct answer is: The series diverges because the \(\displaystyle\lim_{n \to \infty} a_n \) is not 0.

Question **9**

Complete

Not graded

What does the alternating series test say about \(\displaystyle\sum\limits_{n=1}^ \infty (-1)^{n+1} 10^{1/n} \)?

Select one:

- a. Nothing, this is not an alternating series and therefore the alternating series test does not apply.
- b. The series diverges because the \(a_n \) are not decreasing.
- c. The series diverges because the \(\displaystyle \lim_{n \to \infty} a_n \) is not 0.
- d. The series converges.

This is an alternating series

with $a_n = 10^{1/n}$.

Check the alternating series test:

I. $10^{1/n}$ is positive.

II. $10^{1/n}$ is decreasing.

III. $\lim_{n\to\infty} 10^{1/n} = 1$.

The third condition is not met

(the limit $\lim_{n\to\infty} a_n$ must be 0);

therefore, this series diverges

by the alternating series test.

The correct answer is: The series diverges because the \(\displaystyle \lim_{n \to \infty} a_n \) is not 0.

Question **10**

Complete

Not graded

If you estimate

\(\displaystyle\sum\limits_{n=1}^ \infty \frac{(-1)^{n+1}}{(n)^2} \)

by summing the first 99 terms, how many decimal places of accuracy are you guaranteed?

Select one:

- a. 2
- b. 3
- c. 4
- d. 5

If your estimation uses 99 terms, your error is less than a_{100}. Because

$$a_{100} = \frac{1}{100^2} = 0.0001,$$

your error is less than 0.0001. But an error of less than 0.0001 guarantees you 3 decimal places accuracy.

The correct answer is: 3

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